EarthPortal

EarthPortal: curation and FAIRness

Methodology · Results · Proposals

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Where I started

How good are the descriptions of the artefacts in EarthPortal, and how can we improve them?

1

EarthPortal

92 entries in the catalogue: 68 artefacts, 19 views and 5 records without content.

2

Metadata Guidelines

Written by the OntoPortal Federation, being finalised: 99 properties to describe an artefact.

3

O'FAIRe

The FAIRness tool of the OntoPortal portals: 61 questions, 478 points, read from the metadata.

What I was looking for

What?	What is missing in the description of each artefact, according to the Metadata Guidelines?
How much?	How much does it cost in FAIRness, according to O'FAIRe?
Who?	Who can fix it: the author, the curator, or the portal itself?

For every artefact of the catalogue, automatically, and again at each new snapshot.

My methodology in four steps



Collect

The description of every entry, and its O'FAIRe result, from EarthPortal

Score

Where each artefact earns and loses O'FAIRe points

Compare

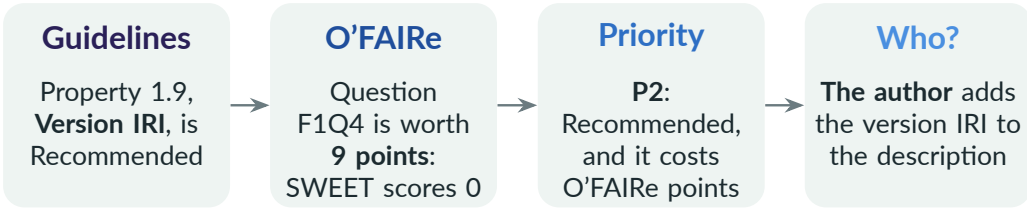
Each description with the 99 properties of the Metadata Guidelines

Audit

One audit per artefact, with a ranked list of actions

Open and reproducible: github.com/EarthPortal/SA_Management · DOI 10.5281/zenodo.23417011

From a missing property to an action



The guidelines decide first; the O'FAIRe points then rank what is left.

Results on the catalogue

67 %

median FAIRness of the 68 artefacts, out of the 386 points that can be earned

90 % | 21 %

of the Mandatory and of the Recommended properties are filled (medians)

95 %

of the 4,378 suggested actions can only be taken by the authors

Where the catalogue loses most points: no DOI or other external identifier (65 of 68 artefacts); few registrations in other repositories; nothing on projects using the artefact or organisations endorsing it (50 of 68).

Latest snapshot of the catalogue: 10 November 2026.

One audit per artefact: SWEET

SWEET: SWEET Ontology

Kind: **artefact** · Snapshot: 2026-11-10 · Portal page: https://earthportal.eu/ontologies/SWEET

This audit has two parts: the FAIRness of the ontology, as measured by O'FAIRe, and its compliance with the Metadata Guideline of the OntoPortal Federation. The suggested actions combine both.

1. FAIRness (O'FAIRe)

O'FAIRe score	Points	Reachable score	Rank
58 / 100	281.75 / 478	73 % of 386	11 of 68 artefacts

Where O'FAIRe points are lost

Question	Points	Lost	O'FAIRe says
F1Q2	0 / 11	11	External identifier is absent.
F1Q4	0 / 9	9	Version URI is absent.
F4Q2	4 / 10	6	The ontology is registered in 2 ontology repositories.
A1Q1	0 / 6	6	The ontology provides 0 resolvable identifier.
A1Q4	0 / 6	6	The ontology is not accessible through a SPARQL endpoint.
R1.2Q2	0 / 6	6	The ontology provides 0 property to inform on its general provenance.
R1.2Q3	0 / 6	6	The ontology provides 0 accrual property.
R1.2Q6	0 / 6	6	The ontology provides 0 property to inform about methodology and tools used.

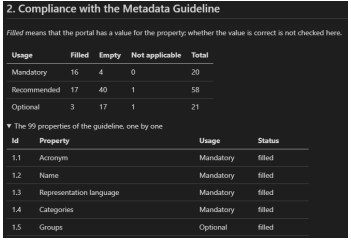
1. FAIRness

O'FAIRe: **58 / 100**
73 % of what can be earned
11th of 68 artefacts

Then each question where SWEET loses points, with O'FAIRe's own sentence.

`audits/SWEET.md` as shown by GitHub on 10 November 2026 (first rows).

One audit per artefact: SWEET



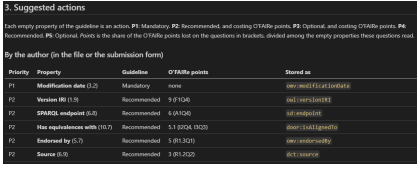
2. Compliance with the Metadata Guideline

Filled means that the portal has a value for the property; whether the value is correct is not checked here.

Usage	Filled	Empty	Not applicable	Total
Mandatory	16	4	0	20
Recommended	17	40	1	58
Optional	3	17	1	21

▼ The 99 properties of the guideline, one by one

Id	Property	Usage	Status
1.1	Acronym	Mandatory	filled
1.2	Name	Mandatory	filled
1.3	Representation language	Mandatory	filled
1.4	Categories	Mandatory	filled
1.5	Groups	Optional	filled



3. Suggested actions

Each empty property of the guideline is an action. **P1**: Mandatory, **P2**: Recommended, and costing OTAFAIRe points. **P3**: Optional, and costing OTAFAIRe points. **P4**: Recommended. **P5**: Optional. *Points* is the share of the OTAFAIRe points lost on the questions in brackets, divided among the empty properties these questions read.

By the author (in the file or the submission form)

Priority	Property	Guideline	OTAFAIRe points	Stored as
P1	**Modification date** (3.2)	Mandatory	none	`oev:modificationDate`
P2	**Version IRI** (1.5)	Recommended	9 (F1Q4)	`owl:versionIRI`
P2	**SPARQL endpoint** (6.0)	Recommended	6 (A1Q4)	`sd:endpoint`
P2	**Has equivalences with** (10.7)	Recommended	5.1 (I2Q4, I3Q3)	`door:isAlignedTo`
P2	**Endorsed by** (5.7)	Recommended	5 (I1.3Q1)	`oev:endorsedBy`
P2	**Source** (8.9)	Recommended	3 (I1.2Q2)	`dct:source`

2. Metadata Guidelines

16 of 20 Mandatory properties filled
17 of 57 Recommended (that apply)

3. Suggested actions

61 actions, ranked: first the missing Mandatory property (Modification date), then the properties that cost points (Version IRI: 9 points).

What I propose

To improve the curation and the FAIRness of the artefacts in EarthPortal

1

Send the audits

Each author receives the audit of their artefact: what to add, and why

2

Curation interface

Built on these audits by Amine Kasmi for the curators (work in progress)

3

Ask at submission

The form does not ask for the licence, the creator or the modification date

4

Repeat

A new snapshot shows what authors and curators have improved

EarthPortal

Thank you

Vielen Dank · Merci beaucoup

Grazie mille

github.com/EarthPortal/SA_Management

Software
10.5281/zenodo.23417011

These slides
10.5281/zenodo.24132003

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